Independent waveform processing and limits to reproducing the Campi Flegrei catalog

Observational reproduction assessment

9 October 2026

Abstract

Tan et al. analyze an observational earthquake catalog; they do not specify a coupled physical forward model. I retrieve original INGV waveform observations and station metadata, execute the explicitly permitted retrained PhaseNet model with independently written preprocessing, and calculate waveform and spectral displays for three templates specified in the supplementary methods. These calculations replace the previous synthetic-only presentation. Three supplementary panel groups are partially reproduced: S1, S2, and S15C. Two additional template figures are related observational analyses; their events differ from the published S15A/B examples. No main-manuscript panel is reproduced. The full association, relocation, magnitude calibration and focal-mechanism workflow remains incomplete. This report separates successful observational processing from missing inputs, restricted derived products, and unimplemented downstream analysis.

1 Source verification and allowed inputs

The local main paper and supplement were re-extracted before any output replacement. Main captions 1–4, supplementary captions S1–S15 and methods S2–S11 are readable. The main PDF emits two font-weight warnings without preventing extraction. Both ZIPs were listed and passed CRC checks; nothing from them was executed or used in analysis.

File	Bytes	Inspection
`tanmodel.pdf`	2140554	Text and captions readable
`tanmodelsupp.pdf`	62164947	Text and captions readable
`science.adw9038_data_s1_to_s3.zip`	12960255	Directory readable; CRC passed; not executed
`science.adw9038_interactive_plot_s1.zip`	8852485	Directory readable; CRC passed; not executed

The data ZIP contains `science.adw9038_data_s1.csv`, `science.adw9038_data_s2.csv`, and `science.adw9038_data_s3.csv`, plus macOS packaging metadata. These are the authors' catalog, associated phases and focal mechanisms. The other ZIP contains `interactive_plot_s1.html` and packaging metadata. The HTML embeds a derived seismicity visualization. These products remain excluded. Exact inventory and SHA-256 checksums are preserved in `docs/file_inventory.json`.

The revised permission allows original observations and station metadata, general PhaseNet and GaMMA tools, and the retrained model artifact. It does not authorize the study's scripts, velocity estimates, catalogs, picks, associations or plotting products. The model repository was inspected as a file tree before downloading only its README, tensor weights, JSON configuration and coordinate-only station CSV. Its fitted velocity models, station-correction file, scripts and output directory were not read. The routine catalog was not downloaded or consulted to select any input window.

2 Actual retrieval and coverage

The INGV landing page failed local TLS issuer verification. The official https://webservices.ingv.it/fdsnws/ station and dataselect endpoints succeeded with normal certificate verification. I requested channel metadata intersecting 21 January 2022–20 March 2025 within 40.7–40.95 degrees N and 13.9–14.3 degrees E, and complete response metadata for CAAM, COLB, CSOB and CSTH. Station epochs describe metadata validity, not guaranteed continuous observation coverage.

I retrieved a calendar-selected validation day, 1 January 2024, at those four stations in hourly three-component chunks. Its date was chosen independently of every event catalog. This is one calendar day within a 1155-day study interval, plus the separate template windows; it does not characterize full-period detection performance. I also retrieved three-minute CAAM/COLB windows around the three template times stated in supplementary S9: 6 May 2024 22:11:10, 8 March 2024 17:40:54 and 17 February 2025 08:06:48 UTC. These method-defined windows support template displays; they do not seed the calendar-day picker.

The temporary-station landing page is accessible. It describes an archive of recordings cut from five seconds before to 55 seconds after author-detected origin times at V0105, V0102, CRUS, CROS, CMDP, CFB1, CFB3, CREM, CSOC, CLAC and CGAU. Its HTTP headers report 3,293,243,207 bytes. Downloading that whole archive would not supply an unbiased continuous record: its selection already incorporates the authors' detection results. It was therefore not ingested into independent discovery. A continuous version would be needed for these stations. RAN's public login page explicitly requires registration/login for SAC or ASCII waveform downloads. No usable account credentials were available; no account was created.

Retrieval URLs, HTTP outcomes, exact sizes and SHA-256 hashes are in `docs/observation_inputs.json`; initial source-access outcomes are in `docs/access_attempts.json`. Two CSOB interruptions occur in the validation day: approximately 9.32 seconds around 17:56 UTC and 8.57 seconds around 22:11 UTC, affecting all three components. They are not interpolated. Actual channel segments, sample counts and time coverage are saved in `data/observations/coverage.csv`. The present run is bounded and does not cover the complete three-year network. Access to other INGV channels is not claimed to be blocked merely because they were not downloaded. A separate FDSN availability-query probe timed out after 25 seconds with no response; support for that endpoint was not established. Therefore the coverage figure uses verified retrieved intervals, not an inferred full archive-availability record.

3 Documented scientific workflow and remaining dependencies

The paper describes retrained PhaseNet detection, GaMMA association, VELEST initial location and velocity calibration, TomoDD differential-time relocation, relative magnitudes, SKHASH/gCAP focal mechanisms and hybrid-template analysis. The methods specify 30-second picker windows, probability threshold 0.3, and exclusion of S picks at CASE, CFB1 and CFB3. Those stations are absent from this validation run. GaMMA requires at least six picks; VELEST applies residual cutoffs of 0.4 then 0.3 seconds and retains events with at least six picks, RMS at most 0.25 seconds and gap at most 300 degrees. TomoDD uses two rounds with picked and waveform-correlated differential times.

Association and relocation were not implemented in this run. The required initial 1D and fixed 3D velocity specifications are not supplied as permitted executable inputs here; the uploaded repository's fitted velocity products remain excluded. The written methods also leave damping, full iteration settings, exact calibration-event selection and correlation weighting incomplete. This is a combination of restricted inputs and unfinished implementation, not a claim that all seismic data are unavailable.

Absolute magnitude estimation presents an additional policy conflict: the paper regularizes against the INGV duration magnitudes, whereas the current instruction permits that catalog only as a final

comparison. I do not import those magnitudes as priors. Relative amplitude differences alone leave a common magnitude offset undetermined. Focal mechanisms further require independent event locations, polarities, amplitude ratios and Green's functions. No synthetic volcano model is introduced to replace these missing analyses. Physical initial or boundary conditions are not applicable to the implemented signal-processing operations.

4 Independent numerical processing

4.1 Waveforms and spectra

ObsPy reads original miniSEED records and StationXML. I reject gaps in the short template streams, check finite samples and 100 Hz sampling, remove the segment mean and divide by the documented channel sensitivity to obtain velocity units. This is sensitivity correction, not full deconvolution of instrument response. The latter is not fully specified for the published display. This choice may affect amplitude and spectral agreement and is not concealed as an exact reproduction.

Each template display spans two seconds before to 18 seconds after its written method time. Each component is normalized by the corresponding CAAM component's maximum absolute amplitude within that window. COLB uses the same CAAM divisor, preserving interstation amplitude ratios. The written caption does not completely specify the normalization window, component pooling or display filter. I display demeaned, sensitivity-corrected signals without an additional bandpass. Spectrograms use 128-sample Hann windows, 112-sample overlap, constant detrending and one-sided power density at 100 Hz. These are explicit display assumptions, not inferred from published pixels.

The frequency index is supplementary equation S4:

$$\mathrm{FI} = \log_{10}\left(\frac{P_{8\text{–}15\,\mathrm{Hz}}}{P_{2\text{–}5\,\mathrm{Hz}}}\right).$$

I calculate it on an 800-sample CAAM vertical window beginning at each method-specified template time. A demeaned rectangular-window DFT gives one-sided Parseval-normalized power; inclusive frequency bins define each band. The exact onset alignment and spectral taper are incompletely specified, so the resulting values are independent measurements for the stated windows, not asserted matches to author event values. The separate 0.2–6 Hz template-matching filter must not be used for an 8–15 Hz FI estimate. An additional independent template search filters the written templates and calendar-day CAAM/COLB vertical traces at 0.2–6 Hz, computes normalized sliding correlations, and averages the two station scores. The scan uses the actual common recorded interval at the two stations, rather than assuming miniSEED blocks begin exactly at the requested hour. Candidates require a score above 0.25. A four-pole causal filter, 8-second minimum peak spacing and exclusion of 10-second chunk edges are explicit implementation choices. Template self-correlation must equal one; insertion at a known offset must recover that offset. Candidates are saved as unreviewed; they are not the published event list. No three-year template search or manual cluster review was completed.

4.2 PhaseNet inference

The explicitly authorized trained tensors are loaded with `weights_only=True`; all state-dictionary keys match the installed SeisBench PhaseNet architecture. The supplied configuration defines ENZ input, 100 Hz sampling, standard-deviation normalization, 1500-sample overlap and 250-sample edge blinding. The installed architecture uses 3001 samples (30.01 seconds), corresponding to the nominal 30-second windows in the written method. The model README specifies a 1 Hz highpass. I use a four-pole causal Butterworth implementation after mean removal; pole count and phase convention are stated implementation assumptions. P and S thresholds are 0.3. Hourly streams are trimmed to requested intervals, split at gaps, and processed without interpolating missing observations. Identical station/phase/peak-time

tuples are deduplicated. Window-edge blinding can miss arrivals near hourly boundaries; there is no completeness claim.

This is new inference using a permitted trained model, not independent training. The saved probabilities are model scores, not calibrated location uncertainties. Pick counts are not earthquake counts and cannot replace Figure 2's seismicity rates. No author picks or events enter preprocessing, thresholds or selection.

5 Results and validation

The run read 102 waveform files containing 312 channel segments. The station map includes 51 study stations. PhaseNet generated 1986 unique picks (295 P and 1691 S). All 13 observational consistency checks passed. The template scan found 0 unreviewed candidate matches across 24 verified hourly intervals (23.813 hours of valid window-start times).

Template	FI	Samples
E1	-0.7862	800
E2	-0.4286	800
E3	-1.0819	800

FI values use the exact windows and spectral choices above. They are not used to tune the workflow. The three FI values span approximately -1.082 to -0.429, within the paper's reported -3.32 to 1.09 range for its full hybrid set. Agreement with that broad range is not event-wise validation or reproduction of temporal evolution. All plotted observational data are saved under `data/observations/`: waveforms in CSV, spectra in compressed NumPy arrays, station coordinates, coverage, picks and hourly pick totals. Plotting reads these saved results and does not read published figure files.

Consistency checks verify CAAM normalization, Parseval power conservation, finite samples, expected sampling and template lengths, model-weight compatibility and probability bounds. The retained independent C++ equation tests validate relative amplitude ratios, signal/noise power, FI on analytic tones, and a stated regularized magnitude least-squares fixture. All 27 checks passed. The maximum absolute FI error across the analytic mixtures was 4e-15; the maximum exact-recovery magnitude error was 1.94e-16. These errors compare against analytic fixtures, not observed earthquakes. An initial template-scan implementation rejected all hours because miniSEED blocks did not begin exactly at requested boundaries. It was corrected to use actual common intervals; a guard now requires all 24 hourly intervals to be scanned and records their valid duration. The initial zero-candidate output was discarded. These tests do not establish detection accuracy or validate association, locations, completeness magnitude or focal mechanisms. There are no independently generated event coordinates against which to compute spatial or depth errors.

6 Reproduction status and discrepancies

No main-manuscript panel is reproduced. S1, S2 and S15C are partially reproduced; all other publication panel groups are not reproduced. The figure-by-figure record below retains every caption-defined main and supplementary group. The machine-readable `docs/panel_record.json` additionally records permitted input files and generation/plot commands. For the three partial groups and two related template examples, commands are `.venv/bin/pythonscripts/analyze_observations.py` and `.venv/bin/pythonscripts/plot_observations.py`. Groups with no numerical execution have null commands. The two related template examples retain their executed commands without claiming reproduction of the corresponding published panels. A written assessment is not counted as numerical

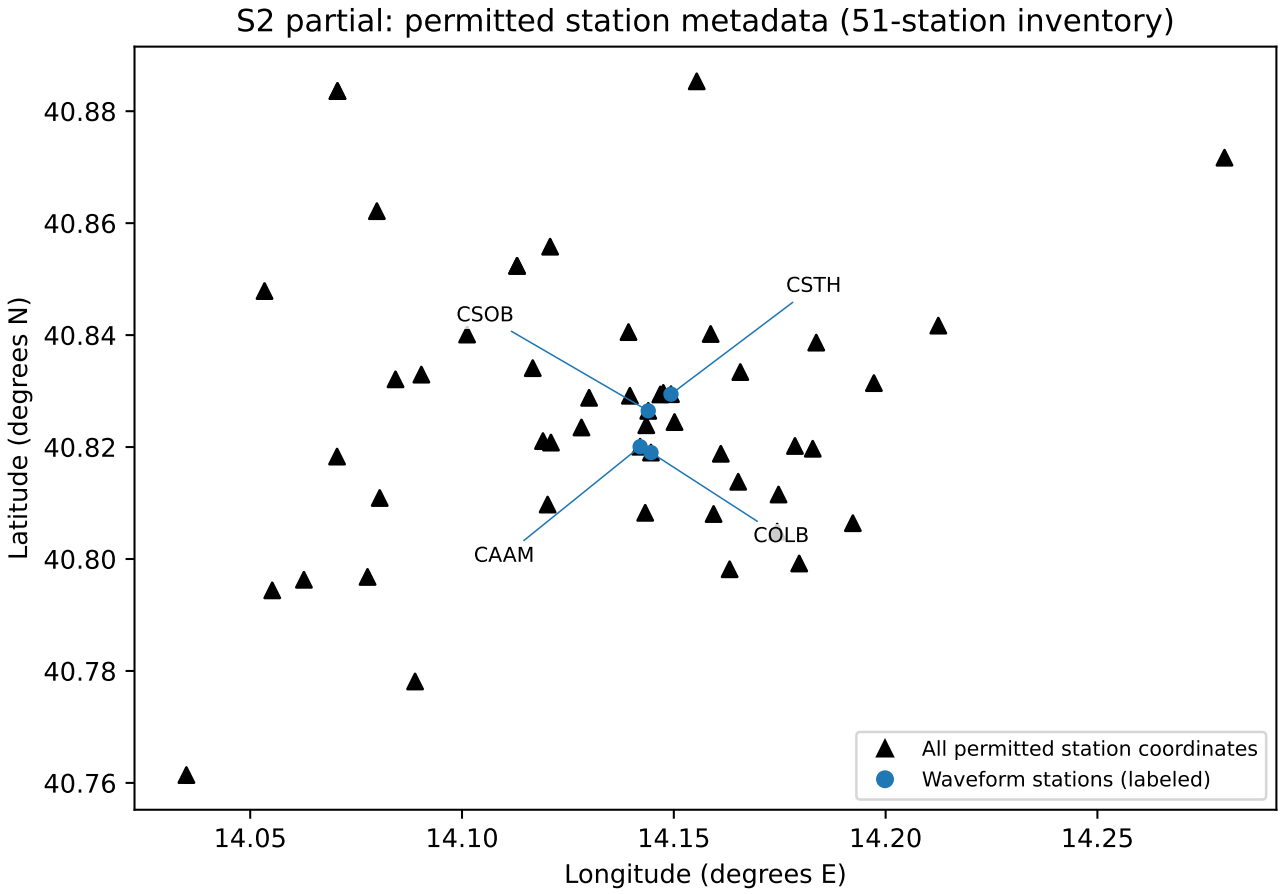


Figure 1: S2, partially reproduced from retrieved station metadata. The permitted coordinate inventory has 51 station names, versus the paper's 48-station description. Official INGV coordinates are used for 22 matching stations; remaining coordinates come from the permitted station metadata CSV. Only the four waveform stations are labeled to avoid overlapping names; all coordinates are saved in CSV, and colocated station names may share symbols. Exact study membership, epochs and the published basemap are not reproduced.

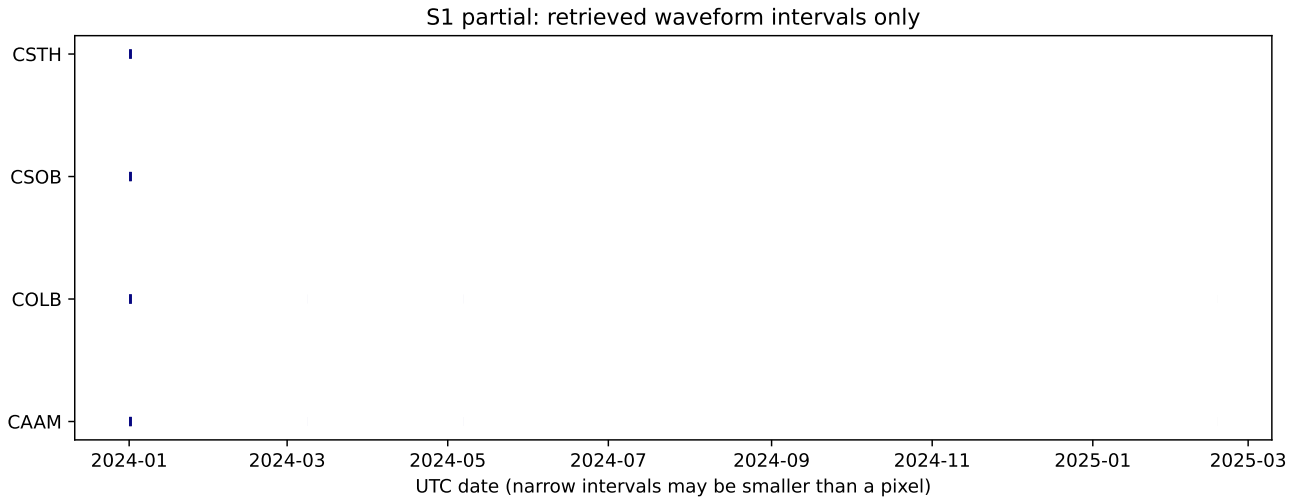


Figure 2: S1, partially reproduced as actual downloaded intervals. Narrow segments occupy less than a pixel at this multiyear scale. This lower bound on retrieved coverage is not the publication's full network-availability record.

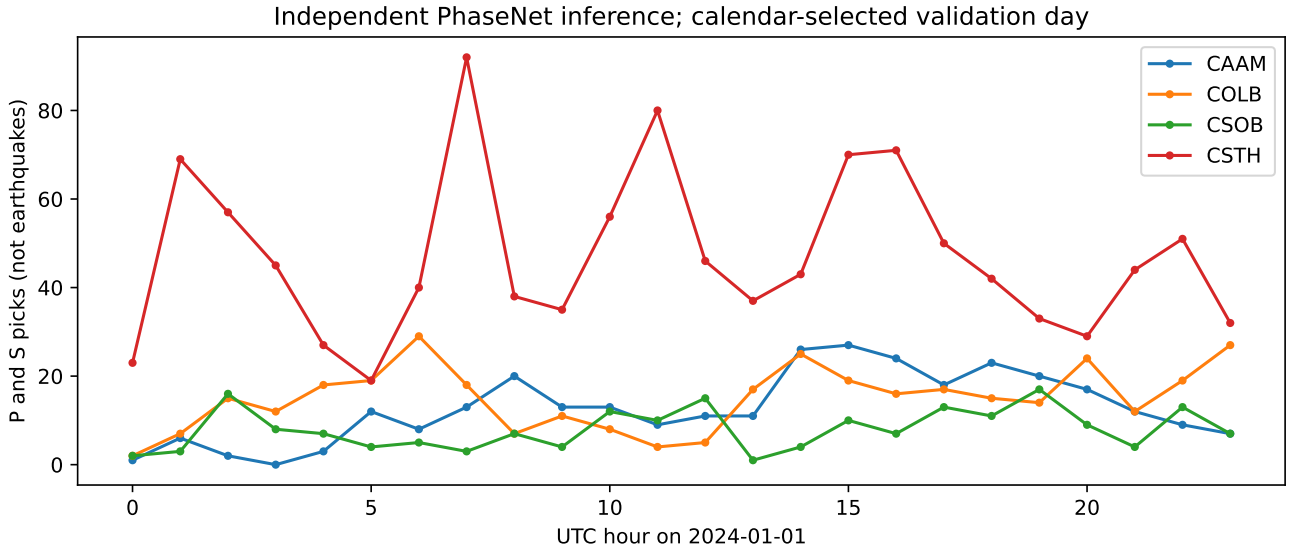


Figure 3: Additional independently generated validation figure. Hourly P/S model picks for a calendar-selected day; these are neither associated earthquakes nor a reproduction of main Figure 2.

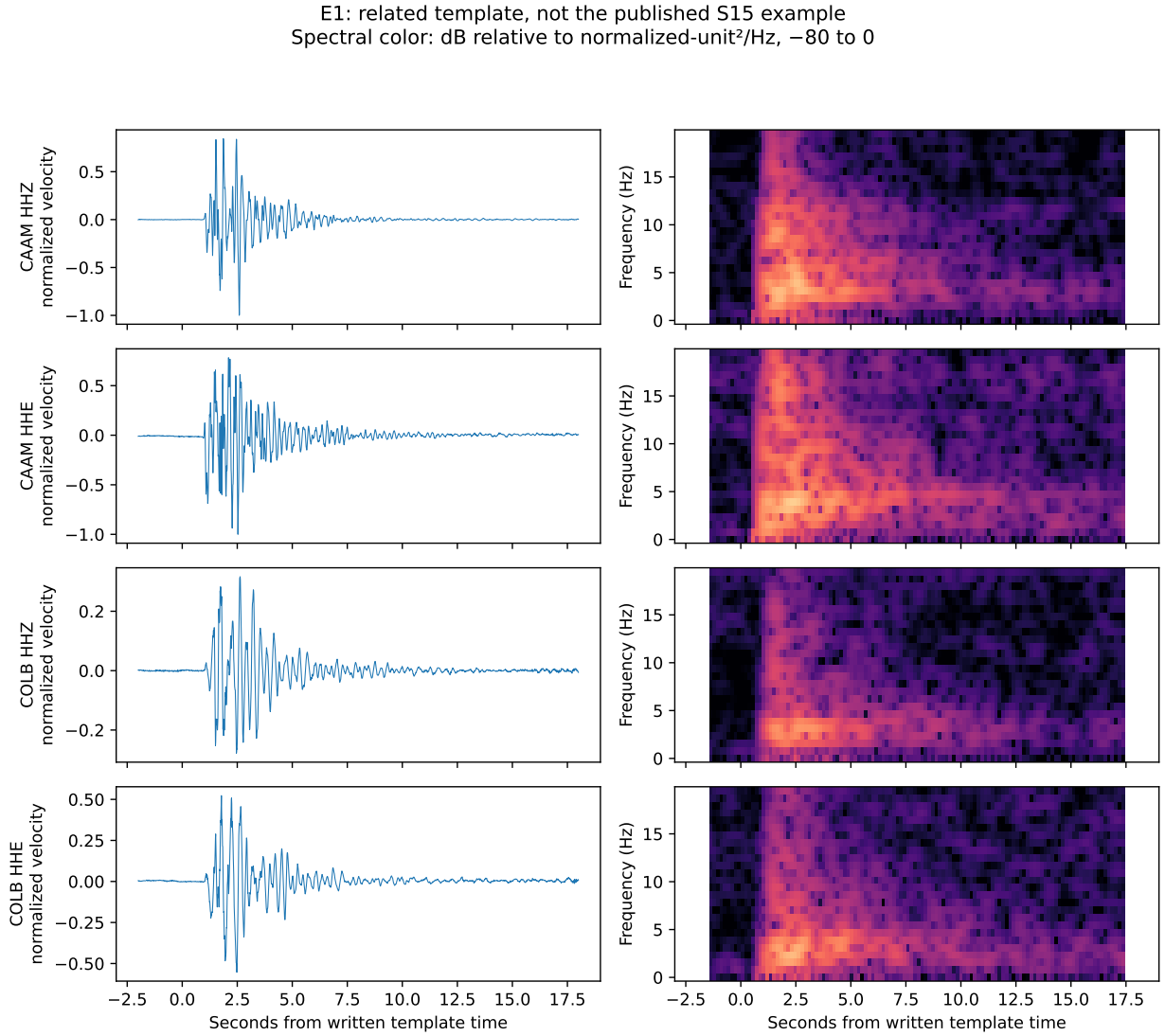


Figure 4: First written-method template: related observational analysis, not reproduction of S15A, which shows a different example. All traces originate from retrieved CAAM/COLB recordings; normalization and spectral assumptions are stated in the methods.

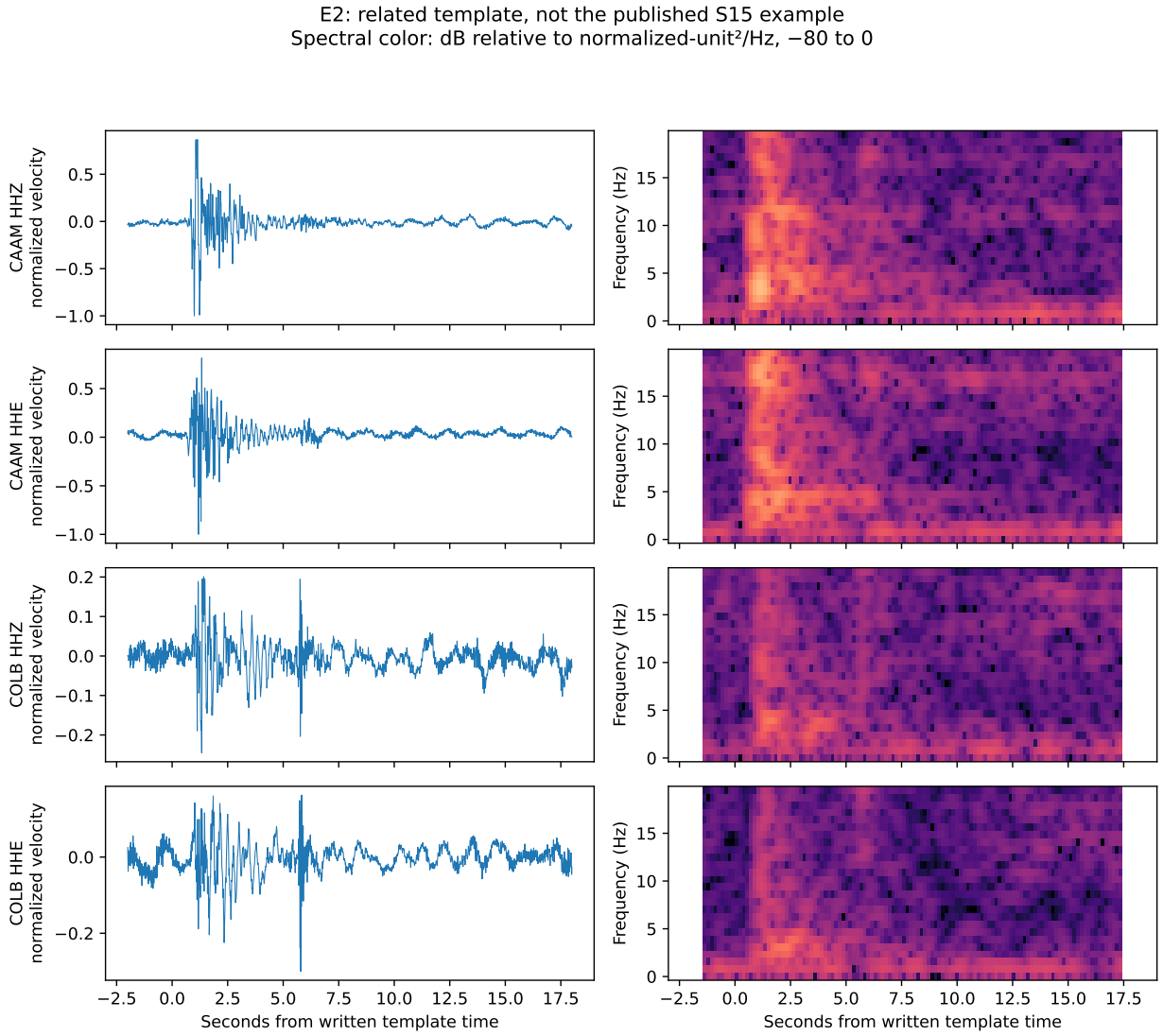


Figure 5: Second written-method template: related observational analysis, not reproduction of S15B, which shows a different example.

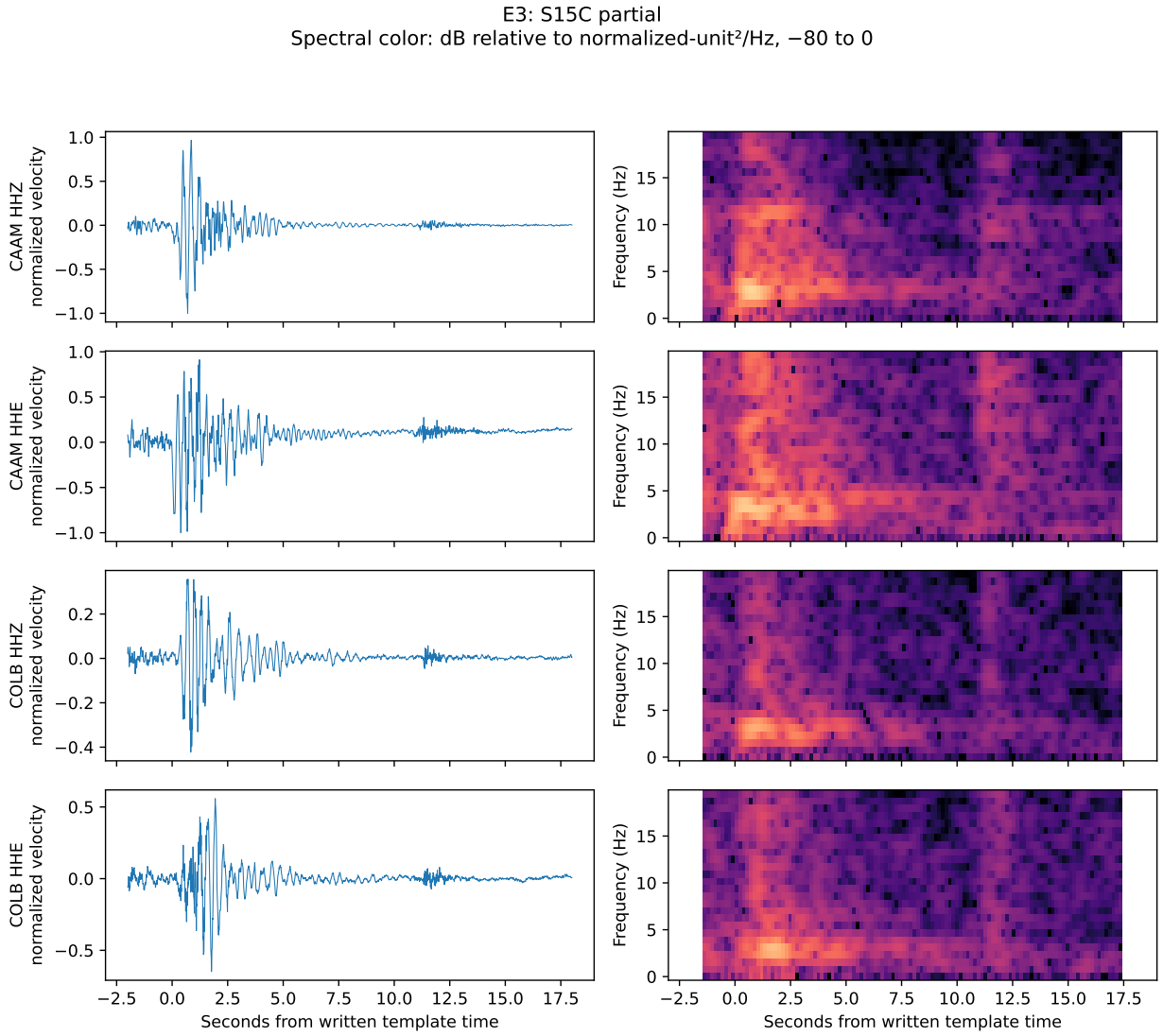


Figure 6: S15C, partial reconstruction for the third written-method template. Exact equivalence to the published example is not asserted.

execution.

The primary shortfall is the missing independent multi-year associated and relocated catalog. Main Figures 1A/C also involve historical catalog information allowed here only as a comparison reference; 1B needs displacement observations not supplied by the seismic services. S3 training examples and S4 original-versus-retrained examples are not reconstructed by applying the retrained model to a different validation day. S14 requires the complete template search and review; S15D's exact shallow example was not independently identified. Plot similarity is not substituted for these missing analyses.

Numerical station coordinates, waveform timing, spectra and FI are available in this run, but no event-matched numerical errors are calculated against prohibited author-derived result files. The report does not interpret absent errors as zero error. After generating the figures, visual comparison with S15 showed that the published A and B examples differ from the written-method E1/E2 template events used here. Their generated plots are therefore related examples, not panel reproductions. E3 corresponds to the S15C example; its dominant low-frequency coda is qualitatively similar, but filtering, normalization, alignment and spectral display differ. No pixel values were extracted or used to revise inputs. The independently generated spectra were compared visually only after generation; the published panels are not embedded in this report. The documented S7 letter inconsistency and S9 strict-versus-inclusive threshold ambiguity remain recorded in `docs/METHODS.md`; neither was silently used to filter an author catalog.

7 Rebuild and replacement

Create the local environment and retrieve inputs with the commands in the README. Run `.venv/bin/pythonscripts/rebuild.py--clean`. The rebuild verifies the four supplied files and downloaded-input hashes, checks readable captions before replacing outputs, compiles and runs the retained C++ consistency checks, executes observational processing and plotting, rebuilds tables, and compiles this report twice. Results and command exit codes are recorded separately from scientific completeness. Retrieval is a separate network step; the cached rebuild is offline.

The previous synthetic method-figure PDFs/PNGs and their plotting script were removed. Previous report source/PDF, panel records, build status and synthetic validation data were regenerated or replaced; the useful C++ validation implementation is retained and clearly labeled. New observational data and figures supersede the synthetic-only presentation. Exact changes are documented in `docs/REPLACEMENTS.md`. Uploaded source files, useful guidance, unrelated work and all Git history are preserved.

Inputs and environment reside in ignored `inputs/` and `.venv/`. Project intermediates and plotting/SeisBench caches reside in `tmp/`. An initial dependency import also created the standard SeisBench configuration cache at `/home/mehffin/.seisbench`; subsequent runs redirect it into the repository. Pip uses its standard user cache under `/home/mehffin/.cache/pip`; no scientific project results reside there. External files need not be committed: their retrieval procedure and checksums accompany the project. The full scientific deliverable remains incomplete despite a successful bounded rebuild.

A Figure-by-figure record

Panel/view	Quantity	Status, specification and limitation
1A	INGV event map and large-event markers	not reproduced. Main p.71, Fig.1 caption. 2005-01-01 to 2025-03-20; origin 14.14 E, 40.82 N; sea-level depth; Md>=4 markers. Routine catalog is allowed only for final comparison, not as plotted analytical input. No independent historical event catalog.
1B	Vertical displacement at Pozzuoli port	not reproduced. Main p.71, Fig.1 caption. 1905-2023; Rione Terra. Historical displacement observations not retrieved; seismic waveform services do not supply this series.
1C	Temporal seismicity and cumulative counts	not reproduced. Main p.71, Fig.1 caption. 2005-2025; cumulative Md>0.2; study interval 2022-01-21 to 2025-03-20. Routine catalog is allowed only for final comparison, not as plotted analytical input. No independent historical event catalog.
2A	Catalog seismicity rates by region	not reproduced. Main p.72, Fig.2 caption. Central vs ring/other regions; 54319 vs 12083 catalog events; time bin unspecified. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
2B	Depth versus time, symbol size by magnitude	not reproduced. Main p.72, Fig.2 caption. Sea-level depth; near-surface 0-0.5 km; central 0.5-2.1; ring sectors to 3.7 km. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
3/map	Epicentral spatial pattern	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
3/northing-depth	Depth versus northing	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
3/ easting-depth	Depth versus easting	not reproduced. Main p.73, Fig.3 caption. Relative origin 14.14 E, 40.82 N; sea-level depths. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
4A	Ring-sector focal mechanisms and seismicity	not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
4B	Central/onshore focal mechanisms and seismicity	not reproduced. Main p.74, Fig.4; supplement S6-S7. High quality: SKHASH A or Kagan angle<30 deg; gCAP preference with stated exception. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.

Panel/view	Quantity	Status, specification and limitation
S1	Station data-availability timeline	partially reproduced. Supplement S12 caption; S2 methods. 48 stations; continuous-data availability. Actual retrieved intervals only; four stations and short windows, not full INGV/IDPC three-year coverage.
S2	Station map	partially reproduced. Supplement S13 caption; S2 methods. 23 broadband, 9 short-period, 16 accelerometers. 51-station permitted metadata inventory differs from the paper's 48-station description; exact epoch/membership and basemap not reproduced.
S3A	Noise example: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3B	Single-event example: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3C	Two overlapping events: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S3D	Five overlapping copies of one event: waveform, phase labels, spectrogram	not reproduced. Supplement S14 caption; S2-S3. 30-s windows; P/S labels; D amplitude rescaling. Exact training traces and labels not identified from permissible metadata. Retraining was not repeated; permitted trained weights used for inference only.
S4A	Swarms at CSOB: waveform, spectrogram, original and retrained picker scores	not reproduced. Supplement S15 caption; S2-S3. 120 s; ~2 h after May 2024 Md4.4 event; phase threshold 0.3. Exact comparison windows and original-network baseline not established. Validation-day picks are not this published example.
S4B	CMIS noise: waveform, spectrogram, original and retrained picker scores	not reproduced. Supplement S15 caption; S2-S3. 1600 s; coastal noise; phase threshold 0.3. Exact comparison windows and original-network baseline not established. Validation-day picks are not this published example.
S5/map	Routine-catalog spatial views; map	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Routine catalog is allowed only for final comparison, not as plotted analytical input. No independent historical event catalog.
S5/northing-depth	Routine-catalog spatial views; northing-depth	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Routine catalog is allowed only for final comparison, not as plotted analytical input. No independent historical event catalog.
S5/ easting-depth	Routine-catalog spatial views; easting-depth	not reproduced. Supplement S16 caption. 12083 events; 1D model and HYPO71. Routine catalog is allowed only for final comparison, not as plotted analytical input. No independent historical event catalog.
S6/map	Initial-location spatial views; map	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.

Panel/view	Quantity	Status, specification and limitation
S6/northing-depth	Initial-location spatial views; northing-depth	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S6/ easting-depth	Initial-location spatial views; easting-depth	not reproduced. Supplement S17 caption. 61809 events; five VELEST runs. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S7A/map	Cluster locations and 2-sigma errors: map and cross-sectional views; map	not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S7A/ easting-depth	Cluster locations and 2-sigma errors: map and cross-sectional views; easting-depth	not reproduced. Supplement S18 caption; S5 methods. Central174, north165, southwest175, east166; HypoDD SVD; Table S1. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S7B	EX uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S7D	EY uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S7F	EZ uncertainty versus differential-pick count	not reproduced. Supplement S18 caption. Log-log axes; NCCP+NCCS. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S7C	EX uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S7E	EY uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S7G	EZ uncertainty density by cluster	not reproduced. Supplement S18 caption. Kernel density estimate; bandwidth unspecified. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S8A	Estimated local versus routine duration magnitude	not reproduced. Supplement S19 caption; S5-S6 methods. Relative-amplitude LS; SNR>=5 dB; >=2 common stations; separation<distance/5. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S8B	Magnitude-frequency distribution and fitted b value	not reproduced. Supplement S19 caption; S6 methods. Published Mc=0.4 by maximum curvature; b=0.97 by ML; bin size unspecified. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.

Panel/view	Quantity	Status, specification and limitation
S9/map	Filtered-catalog spatial views and feature labels A-D; map	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S9/northing-depth	Filtered-catalog spatial views and feature labels A-D; northing-depth	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S9/ easting-depth	Filtered-catalog spatial views and feature labels A-D; easting-depth	not reproduced. Supplement S20 caption. 38103 events; NCCP+NCCS threshold 100. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S10/2022	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2022. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S10/2023	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2023. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S10/2024	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2024. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S10/2025	Annual seismicity map	not reproduced. Supplement S21 caption. Year 2025 through March20. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S11/swarm-1	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S11/swarm-2	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S11/swarm-3	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S11/swarm-4	Map of one of four major swarms	not reproduced. Supplement S22 caption. 2022-2025; exact swarm windows not enumerated in caption. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S12A	Ring-sector division	not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.

Panel/view	Quantity	Status, specification and limitation
S12B	Ring-sector division in complementary spatial view	not reproduced. Supplement S23 caption; S8. East, southwest, north; sector boundaries by location/depth. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S12C	Common-receiver gather: east / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S12D	Common-receiver gather: southwest / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S12E	Common-receiver gather: north / HHZ	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S12F	Common-receiver gather: east / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S12G	Common-receiver gather: southwest / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S12H	Common-receiver gather: north / HHN	not reproduced. Supplement S23 caption. CPOZ; -1 to +7 s about P; >4 Hz; north sector ordered by easting, others northing. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S13A	Full caldera seismicity on satellite reference	not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S13B	Onshore seismicity on satellite reference	not reproduced. Supplement S24 caption. Depth-coded relocated events; Google Earth basemap. No independent associated and relocated multi-year event catalog; author-derived catalogs and velocity estimates excluded. Downstream inversion not implemented.
S14/1	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. Full three-year template search and manual review not performed. Three method-specified templates alone do not reproduce cluster history.
S14/2	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. Full three-year template search and manual review not performed. Three method-specified templates alone do not reproduce cluster history.

Panel/view	Quantity	Status, specification and limitation
S14/3	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. Full three-year template search and manual review not performed. Three method-specified templates alone do not reproduce cluster history.
S14/ unclustered	CAAM waveforms and event annotations	not reproduced. Supplement S25 caption; S9-S11. 1-10 Hz display; 8-s FI at 2-5 and 8-15 Hz; FI<0 colored red. Full three-year template search and manual review not performed. Three method-specified templates alone do not reproduce cluster history.
S15A	Hybrid family 1: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel.
S15B	Hybrid family 2: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Post-generation visual comparison shows a different published example from the written-method template used here. The generated family-template figure is related analysis, not reproduction of this panel.
S15C	Hybrid family 3: waveforms and spectrograms	partially reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Written template times used; exact published example alignment, display filtering and spectral settings incompletely specified. Sensitivity-corrected velocities and stated independent display settings used.
S15D	Shallow volcano-tectonic example: waveforms and spectrograms	not reproduced. Supplement S26 caption; S9-S10. CAAM/COLB HHZ/HHE; amplitudes normalized by CAAM maximum. Exact shallow VT example not identified by written method alone; no author catalog or plot values used to select it.

B Equation consistency checks

Check	Absolute error	Tolerance
equal_amplitudes	0	1e-14
decade_amplitude	0	1e-14
ratio_reversal	0	1e-14
common_station_gain	5.55e-17	1e-14
zero_amplitude_rejected	0	0
negative_power_rejected	0	0
fi_mixture_0	4e-15	1e-10
fi_mixture_1	8.88e-16	1e-10
fi_mixture_2	1.78e-15	1e-10
fi_mixture_3	1.55e-15	1e-10
fi_mixture_4	1.78e-15	1e-10
fi_mixture_5	2.11e-15	1e-10
fi_mixture_6	2.22e-16	1e-10

Check	Absolute error	Tolerance
fi_mixture_7	2.11e-15	1e-10
fi_mixture_8	1.33e-15	1e-10
fi_mixture_9	8.88e-16	1e-10
fi_mixture_10	1.33e-15	1e-10
fi_mixture_11	4.44e-16	1e-10
fi_mixture_12	1.78e-15	1e-10
parseval	3.33e-15	1e-10
fi_circular_shift	7.22e-16	1e-10
fi_gain	8.88e-16	1e-10
snr_5s_2s_windows	8.88e-16	1e-10
magnitude_exact_recovery	1.94e-16	1e-12
magnitude_pair_order	0	1e-12
magnitude_gauge_shift	4.44e-16	1e-12
unanchored_problem_rejected	0	0

C Sources

Tan et al. (2025), *A clearer view of the current phase of unrest at Campi Flegrei caldera*, Science 390, 70–75: local `tanmodel.pdf` and `tanmodelsupp.pdf`. Raw waveforms and station metadata: INGV FDSN web services. Permitted trained artifact and written usage notes: https://github.com/iamharrison2000/Campi-Flegrei-Earthquake-Monitoring, pinned commit in the input manifest. Temporary-record source: https://dataserver.ov.ingv.it/users/anna.tramelli/seismic_records/index.html. RAN access requirements: https://ran.protezionecivile.it/IT/autenticazione.php. Independent scripts use ObsPy and SeisBench; installed versions are pinned in `config/requirements-observations.txt`.